

Self-Interacting Stochastic Dynamics with Exponential Memory: Markov Embedding and a Linear Memory-Cloud Baseline

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We study a point-valued stochastic process coupled to a self-generated spatial memory density with exponential forgetting. The visible trajectory is generally non-Markovian, whereas the trajectory together with the memory field is Markov. For the normalized memory update we derive an exact recursion for the memory centroid and, after a local linearization of the self-induced force, an autoregressive relative mode with an explicit stationary radius. This mode supplies a discriminating null hypothesis for apparent localization: in nine active long-run slices, spanning ambient dimensions $d = 3$ to 20 and run lengths up to 3×10^8 updates, the measured co-moving radius agrees with the finite-memory linear prediction to a median relative error of 0.76% and a maximum of 1.15%. Matched one- and two-scale kernels also collapse when parametrized by their local restoring curvature. The current scalar evidence therefore supports a reproducible co-moving relaxation cloud, but does not isolate a nonlinear metastable state or phase transition. We formulate the stronger diagnostics that such a claim would have to pass.

I. INTRODUCTION

Self-interacting stochastic processes describe motion whose future drift depends on its own past. Established examples include reinforced walks, true self-avoiding or self-repelling motion, Brownian polymers, and self-interacting diffusions [1–5]. In the diffusion framework of Benaïm, Ledoux, and Raimond, the interaction is driven by a normalized occupation measure; for symmetric interactions, Benaïm and Raimond proved convergence of that measure toward the critical set of a nonlinear free-energy functional [6, 7]. Herrmann and Roynette instead studied self-attracting diffusions driven by an unnormalized integral over the full past and established convergence or boundedness in specific one-dimensional cases [8]. Neither construction uses exponential forgetting.

The most direct recent comparison is the aging-kernel model of Milišić, Meunier, and Roux, which treats linear interactions with weighted history and includes an explicitly solvable exponential-memory case [9]. Exponential kernels also permit Markovian embeddings in generalized Langevin and other memory equations by adding auxiliary variables [10–12].

The contribution here is deliberately more specific than a claim of a new general memory principle. We combine a discrete point process with a deposited spatial memory field and a potentially nonlinear interaction kernel, make the augmented Markov state explicit, and derive a linear center-relative null model against which localized simulation output can be tested. The last step changes the interpretation of the numerical evidence: the compact scalar memory clouds are real and reproducible, but their measured size is almost completely explained by the linear finite-memory mode.

II. MODEL AND MARKOV EMBEDDING

Let $\Omega \subseteq \mathbb{R}^d$ carry Lebesgue measure. Let $G_\sigma \in L^1(\Omega)$ be nonnegative and normalized, and suppose that $K(x, y)$ is measurable and continuously differentiable in x , with $\nabla_x K(x, \cdot)$ integrable against the finite memory measures considered below. Boundary or extension conventions are chosen so that each update remains in Ω and deposition preserves its stated mass.

At update n , the augmented state is

$$z_n = (x_n, \rho_n), \quad (1)$$

where $x_n \in \Omega$ and ρ_n is a nonnegative memory density. The update is

$$x_{n+1} = x_n + \varepsilon \xi_n - \eta \nabla \Phi_n(x_n), \quad (2)$$

$$\rho_{n+1}(x) = (1 - \lambda_m) \rho_n(x) + \beta G_\sigma(x - x_{n+1}), \quad (3)$$

$$\Phi_n(x) = \int_{\Omega} K(x, y) \rho_n(y) dy, \quad (4)$$

where $0 < \lambda_m < 1$, $\beta, \varepsilon, \eta \geq 0$, and ξ_n are independent centered noise vectors with unit covariance. The gradient in Eq. (2) acts on the first kernel argument:

$$\nabla \Phi_n(x) = \int_{\Omega} \nabla_x K(x, y) \rho_n(y) dy. \quad (5)$$

The total memory mass $M_n = \int_{\Omega} \rho_n(x) dx$ obeys

$$M_{n+1} = (1 - \lambda_m) M_n + \beta. \quad (6)$$

Thus unit mass is preserved when $M_0 = 1$ and $\beta = \lambda_m$. We use this normalized convention below and write $\lambda_m = \beta = \alpha$. Normalization fixes the memory scale; it is not a probability interpretation of the dynamics.

Given z_n and a fresh noise draw, Eqs. (2) and (3) determine the law of z_{n+1} . Hence $\{z_n\}_{n \geq 0}$ is a time-homogeneous Markov process. The visible process

$\{x_n\}_{n \geq 0}$ is generally not Markov because two equal positions with different memory fields can have different next-step drifts. Equivalently, the transition kernel $P(z, dz')$ acts on observables through

$$(Uf)(z) = \int f(z')P(z, dz') = \mathbb{E}[f(z_{n+1}) | z_n = z], \quad (7)$$

a positive unital Markov operator with $U^{m+n} = U^m U^n$ [13, 14].

The one-step affine memory map is algebraically invertible if x_{n+1} is known. What is lost is the ordered history represented by the current field: different histories can yield the same ρ_n . We therefore use “history compression,” not thermodynamic irreversibility, for the direction selected by the forward Markov semigroup.

III. EXPONENTIAL MEMORY AND EFFECTIVE TIME

Iterating the normalized memory update gives, for $n > k$,

$$\rho_n(x) = (1 - \alpha)^{n-k} \rho_k(x) + \alpha \sum_{j=k+1}^n (1 - \alpha)^{n-j} G_\sigma(x - x_j). \quad (8)$$

A contribution of age m therefore has weight $\alpha(1 - \alpha)^m \simeq \alpha e^{-\alpha m}$ for small α . The memory-persistence scale is $O(\alpha^{-1})$ updates.

The update index is not physical time. When observables vary slowly across one persistence window, the dimensionless parameter $t = \alpha n$ is nevertheless a useful comparison scale. Under a joint small-step scaling with $D = \lim \varepsilon^2 / (2\alpha)$ and $\gamma = \lim \eta / \alpha$, the visible update has the formal effective form

$$dX_t = -\gamma \nabla \Phi[\rho_t](X_t) dt + \sqrt{2D} dW_t. \quad (9)$$

This is a scaling ansatz, not a convergence theorem.

IV. LINEAR MEMORY-CLOUD NULL MODEL

In the normalized convention, assume the deposition kernel is centered and define the memory centroid

$$m_n = \int_{\Omega} y \rho_n(y) dy. \quad (10)$$

Equation (3) then gives the exact recursion

$$m_{n+1} = (1 - \alpha)m_n + \alpha x_{n+1}. \quad (11)$$

Suppose the force is sampled inside a region where its center-relative part is well approximated by an isotropic linear restoring term,

$$-\eta \nabla \Phi_n(x_n) \simeq -g(x_n - m_n). \quad (12)$$

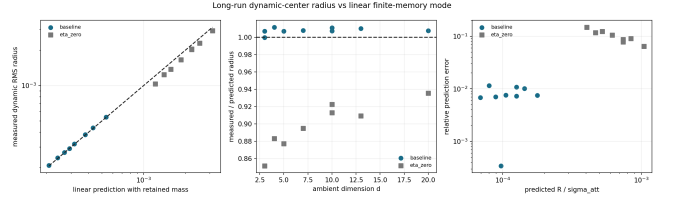


FIG. 1. Measured co-moving radii compared with the finite-memory linear prediction. Active slices remain close to the diagonal across ambient dimension and run length. The $\eta = 0$ controls are shown separately because the active-force formula is not an exact identity for their finite-window trace estimator.

For a smooth translation-invariant kernel, g is set by the local curvature, the retained memory mass, and η . Writing $r_n = x_n - m_n$, Eqs. (2) and (11) yield

$$r_{n+1} = (1 - \alpha)(1 - g)r_n + (1 - \alpha)\varepsilon\xi_n. \quad (13)$$

The translation mode has eigenvalue one, while the relative mode has the real multiplier $(1 - \alpha)(1 - g)$. If its modulus is below one, its stationary Euclidean root-mean-square radius is

$$R_{\text{lin}} = \frac{\sqrt{d}(1 - \alpha)\varepsilon}{\sqrt{1 - (1 - \alpha)^2(1 - g)^2}}. \quad (14)$$

For a truncated history buffer, the implemented benchmark replaces the infinite-memory mass in g by the retained mass $M_{\text{stored}} = 1 - (1 - \alpha)^H$.

Equation (14) is the relevant null hypothesis for compact co-moving clouds. Localization relative to m_n is not, by itself, evidence for a nonlinear metastable state: it is already produced by the stable linear relative mode.

V. NUMERICAL TEST

The current reference implementation was tested in nine active long-run slices. Each slice uses five seeds; run lengths are $N = 3 \times 10^7$ updates, with one $N = 3 \times 10^8$ check, and ambient dimensions range from $d = 3$ to 20. Across these slices, the median relative difference between the measured co-moving radius and Eq. (14) is 0.76% and the maximum is 1.15%. Figure 1 shows the reconciliation.

Two additional controls constrain the interpretation. First, an attractive single-scale kernel and the earlier two-scale kernel produce numerically indistinguishable scalar observables when matched by their local restoring curvature. The repulsive component is therefore not identified by the compact branch in the sampled Taylor regime. Second, increasing the predicted radius to $R_{\text{lin}}/L = 0.3$ produces a seed-stable 6.2% superlinear radius correction but no corresponding memory-shape transition. The pre-registered composite decision remains inconclusive, while its residence-based supports are either scale-dependent or saturated in the $\eta = 0$ control.

The covariance participation dimension of an isotropic cloud approaches the supplied ambient dimension. Consequently, a value near three in a three-dimensional embedding is an expected shape check, not evidence for dimension selection, and is not used as a primary result here.

VI. CRITERIA BEYOND THE NULL MODEL

We use *localized memory cloud* for the descriptive state established above. The project term *dynamical knot* is reserved for a stronger finding: a nonlinear metastable regime that survives comparison with Eq. (13) and matched no-feedback controls. For a region $\mathcal{K} \subset \Omega$, one possible observable is

$$M_n(\mathcal{K}) = \int_{\mathcal{K}} \rho_n(x) dx. \quad (15)$$

However, elevated or long-lived $M_n(\mathcal{K})$ is discriminating only when the region definition scales with the cloud and the same statistic is not saturated under the null.

A stronger test uses reduced features of the augmented state and estimates a lagged transfer operator. Almost-invariant sets or nontrivial slow modes can then be compared across lag, discretization, seeds, and matched controls [15–17]. For an eigenvalue $\mu_i(\tau)$, the implied rate is

$$\Gamma_i(\tau) = -\frac{1}{\tau} \log |\mu_i(\tau)|. \quad (16)$$

A local frozen-memory Hessian gives an OU approximation to such relaxation, but it does not establish metastability by itself. When a relaxation rate is independently

identified, the dimensionless ratio

$$\hat{\Gamma} = \frac{\sigma^2}{D} \Gamma_{\text{rel}} \quad (17)$$

compares relaxation with diffusion on the deposition scale. No physical mass interpretation follows without an additional operational calibration.

VII. DISCUSSION

The model provides a transparent field-valued Markov embedding of a self-interacting trajectory with exponential forgetting. Its main present numerical result is not a nonlinear knot claim. It is the more discriminating finding that the scalar compact branch is a co-moving relaxation cloud quantitatively controlled by the exact linear center-relative mode over the tested long-run slices.

This negative separation is useful: future mechanisms must produce an observable that the linear mode and matched controls cannot reproduce. Candidate tests include a scale-aware almost-invariant set, a reproducible nonlinear bifurcation, or a relational response channel with independently evolved source and target states. The current equations do not establish physical spacetime, finite signal speed, dimension selection, quantum dynamics, particle structure, or calibrated mass.

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